

Framingham Hard CHD Risk Score

Study notes · Medical background, model, implementation and self-test

Learning goals. After these notes you can (1) explain what the score predicts and why each risk factor matters medically, (2) explain the Cox formula in plain words, (3) name the limits of the score, and (4) answer the self-test questions in your own words.

1. What the score predicts

The score estimates the **10-year risk of hard coronary heart disease (CHD)**: a **heart attack (myocardial infarction)** or **death from coronary disease**. "Hard" means only clear, measurable events. Angina (chest pain) and stroke are **not** counted.

The disease behind it: atherosclerosis

Cholesterol (mainly LDL) is deposited in the walls of the coronary arteries and forms plaques. The artery narrows and the heart muscle gets less oxygen. If a plaque ruptures, a blood clot forms and can block the artery completely: that is the heart attack.

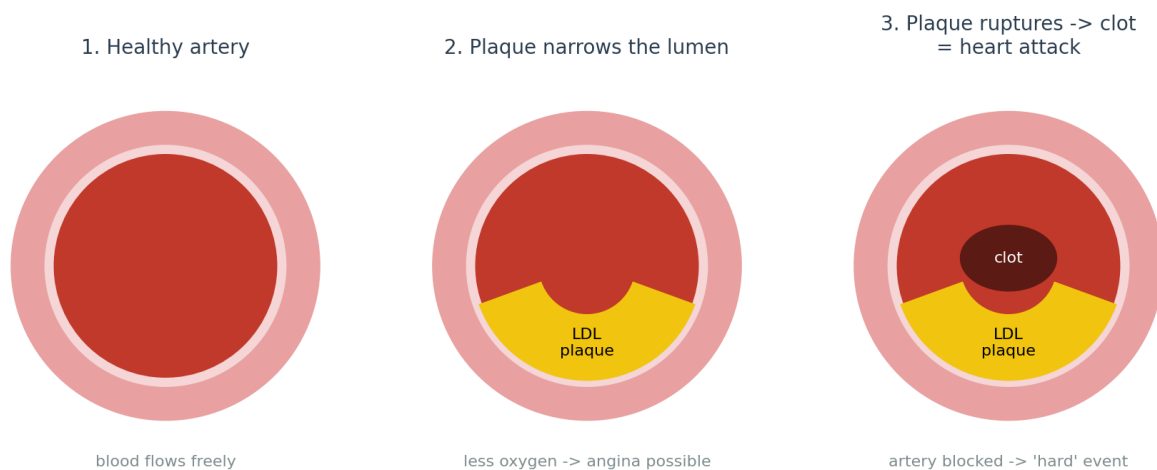


Figure 1 · From healthy artery to heart attack. Every risk factor in the score speeds up this process.

Who the score is for: ages 30–79, **primary prevention** (no CHD yet), **no diabetes**. Diabetics are excluded because ATP III treats diabetes as a *CHD risk equivalent*: they are high-risk anyway.

2. The six predictors

Predictor	Effect	Medical reason
Age	↑ risk	The strongest factor. Vessels accumulate damage over decades.
Total cholesterol	↑ risk	More cholesterol in the blood means more material for plaques.
HDL cholesterol	↓ risk (negative coefficient)	"Good" cholesterol: carries cholesterol away from the vessel walls back to the liver.
Systolic BP	↑ risk	High pressure damages the inner vessel wall (endothelium); plaques form there more easily.
Treated hypertension	↑ risk	Medication lowers today's value but can't undo years of damage (see Figure 2). Counts only if SBP > 120.
Smoking	↑ risk	Damages the endothelium, promotes inflammation and blood clotting.

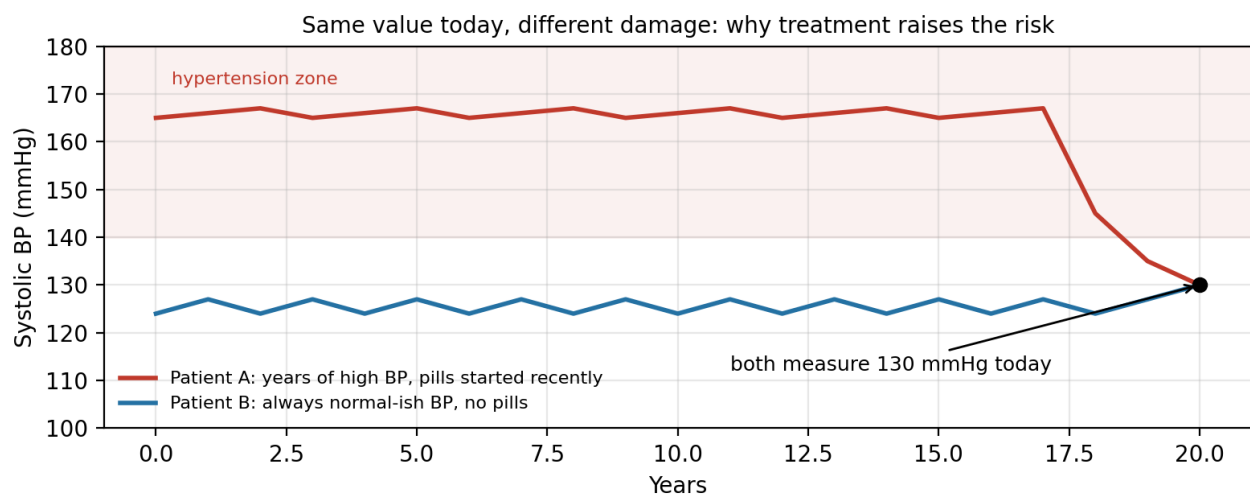


Figure 2 · Both patients measure 130 mmHg today, but patient A carries years of damage. That is why treatment raises the risk at the same value.

Interaction terms (typical exam question)

Age × cholesterol is negative: high cholesterol matters relatively less in old age (survivor effect: people with very harmful levels often had their event earlier). **Age × smoking is negative:** the relative extra risk of smoking decreases with age. The **age cap** (ln 70 for men, ln 78 for women) exists because the study had too few very old smokers.

3. The math: a Cox proportional hazards model

$$\text{Risk} = 1 - S_0 \wedge \exp(\Sigma \beta x - \Sigma \beta x_{\text{mean}})$$

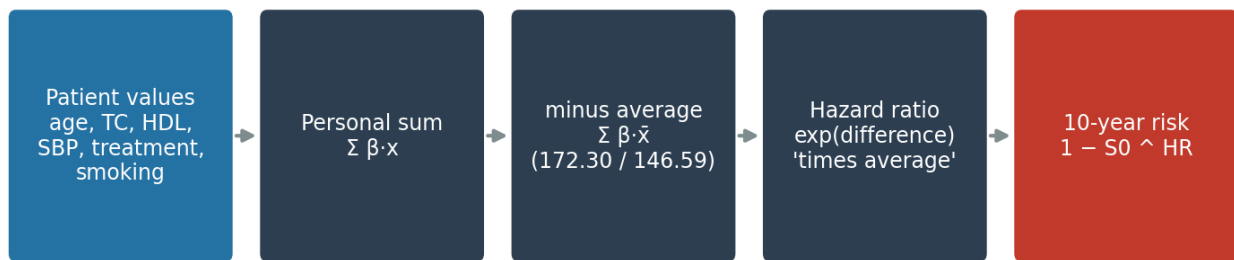


Figure 3 · From patient values to risk in five steps.

Symbol	Meaning in plain words
S_0 (baseline survival)	Probability that an average participant has no event in 10 years. Men 0.9402 (average risk $\approx$ 6 %), women 0.98767.
$\Sigma \beta x$	Your personal sum of coefficient $\times$ value.
$\Sigma \beta x_{\text{mean}}$	The same sum for the average participant: 172.30 (men), 146.59 (women).
$\exp(\text{difference})$	The hazard ratio : how many times riskier you are than the average. HR = 1 means exactly average.
$\ln(\dots)$	Risk grows proportionally, not linearly, so values enter as logarithms.

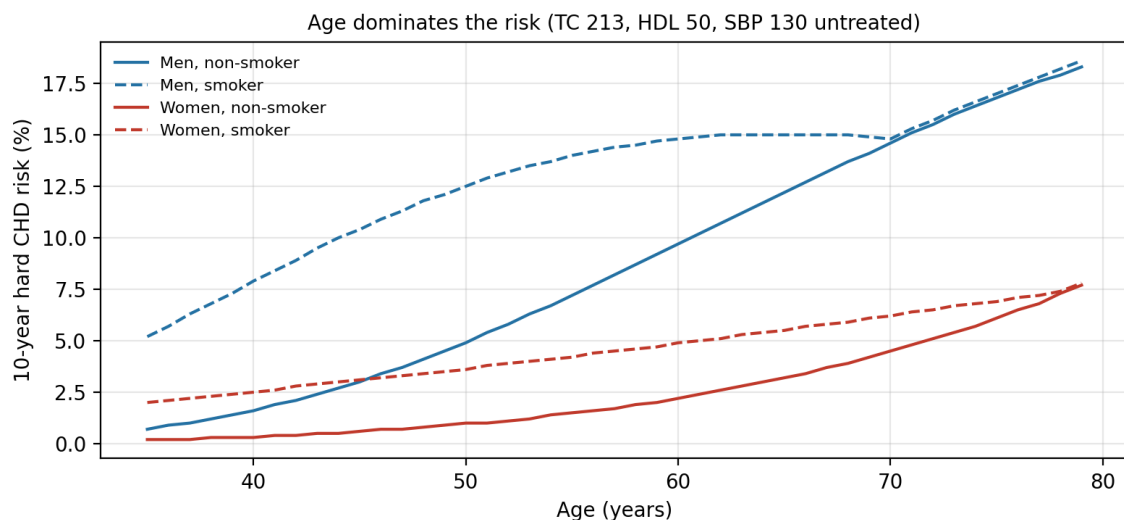


Figure 4 · Output of your own risk.py: age dominates, smoking roughly doubles the risk in middle age.

4. Limitations: think critically

- **Population:** developed on a US cohort (Framingham, Massachusetts, mostly white). In Europe it tends to **overestimate** risk. In Germany the standard is **SCORE2** (ESC 2021); the PARS chart is calibrated for Iran.
- **Population estimate, not a prediction:** "Of 100 people like you, about 7 will have an event in 10 years", not whether *this* person will.
- **Units:** the formula expects mg/dL. Cholesterol conversion: **mmol/L × 38.67 = mg/dL**.

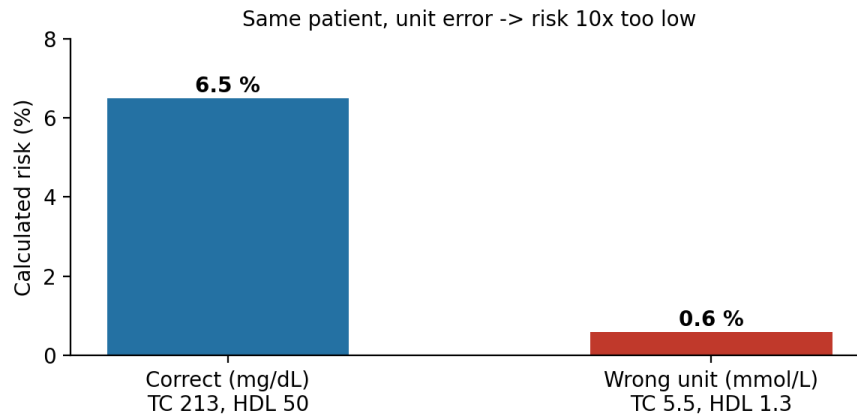


Figure 5 · Unit error in your own code: 6.5 % becomes 0.6 %. Wrong but plausible-looking, the most dangerous kind of error.

5. The informatics side

What you did	Why it matters
Validated every coefficient against the Framingham Heart Study	Primary source, not a blog
Automated tests (reference values, rules, plausibility, invalid input)	Errors are caught every time the code changes
Input validation (age 30–79, plausibility ranges)	In medicine, a wrong number is worse than no number
Label "Demo, not a diagnostic tool"	Real clinical decision software can be a medical device in the EU

6. Self-test

Cover the right column and answer in your own words. Repeat until all seven are clean.

#	Question	Model answer
1	What does "hard CHD" mean, and what does it exclude?	Only a heart attack or death from coronary disease . Angina and stroke are not counted.
2	Why is the HDL coefficient negative?	HDL carries cholesterol away from the vessel walls back to the liver: protective.
3	Why does <i>treated</i> hypertension increase the risk?	Medication lowers today's value but can't undo years of damage . At the same BP, the treated patient has the higher risk.
4	Why are diabetics excluded?	Diabetes is a CHD risk equivalent : they are high-risk without needing a score.
5	What does S_0 mean in plain words?	Probability that an average participant has no event in 10 years (men 94 % → average risk ≈ 6 %).
6	Why not use Framingham directly for German patients?	Developed on a US population ; in Europe it overestimates . Use SCORE2 .
7	What if cholesterol comes in mmol/L unconverted?	Values ≈ 40× too small → risk far too low (6.5 % → 0.6 %). High-risk patient looks safe. Hence plausibility ranges .

Question	1	2	3	4	5	6	7
First try (25.09.2026)	⚠	✓	⚠	✓	✗	⚠	⚠
Second try							
Third try							